

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Time: 25 min.

SET B
Subject: Physics [1011]

Choose the best option among the four options on your answer sheet.

Choose the best alternative.

GROUP 'A' [11×1=11]

Multiple choice questions

1. Turpentine oil is flowing through a tube of length 'l' and radius 'r'. The pressure difference between the two ends of the tube is P. The viscosity of the oil is given by: $\eta = \frac{P(r^2 - x^2)}{4vl}$, where 'v' is the velocity of oil at a distance x from the axis of the tube. The dimensions of η are:

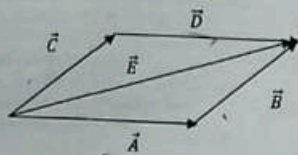
- a) $[M^0 L^0 T^0]$ b) $[M^1 L^1 T^{-1}]$ c) $[M^1 L^2 T^{-2}]$ d) $[M^1 L^{-1} T^{-1}]$

2. The expression $\left[\left(\frac{1}{\sqrt{3}}\hat{i} - \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}\right)\right]$ is a:

- a) Null vector b) Unit vector
c) Vector of magnitude $\sqrt{2}$ d) Scalar

3. From the figure:

- a) $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$
b) $\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} = \vec{0}$
c) $\vec{A} + \vec{B} + \vec{C} + \vec{D} - \vec{E} = \vec{0}$
d) $\vec{A} + \vec{B} + \vec{C} + \vec{D} - 2\vec{E} = \vec{0}$



4. What is the angle between $(\vec{P} + \vec{Q})$ and $(2\vec{P} + 2\vec{Q})$?

- a) Zero b) π c) $\frac{\pi}{2}$ d) π

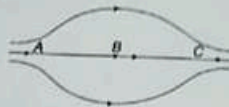
5. An aeroplane is moving in a circular path with a speed of 125 Km/hr; what is the change in velocity in half revolution?

- a. 500 Km/hr b. 250 Km/hr c. 125 Km/hr d. Zero:

6. A certain vector in x-y plane has an x-component of 12 m and y-component of 8m. It is then, rotated in xy plane so that its y-component is halved. Then its new x component is approximately

- a) 14m b) 10.12m c) 2.14 m d) 13.93 m

7. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests:



- a) $E_A = E_C > E_B$
b) $E_A = E_C < E_B$
c) $E_A > E_B > E_C$
d) $E_A = E_B = E_C$

8. An isolated conducting sphere is given a negative charge. Its mass

- a) gets increased b) gets decreased
c) remains the same d) mass is not involved during electrification

9. The magnitude of electric field strength E such that an electron placed in it would experience an electrical force equal to its weight, is given by:

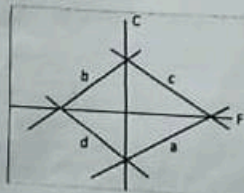
- a) mg b) mg/e c) mg/e² d) e/mg

10. A temperature degree on Kelvin Scale is same as:

- a) a temperature degree on Celsius Scale
b) a temperature degree on Fahrenheit Scale
c) a temperature degree on Rankine Scale
d) 1/32th part of temperature degree on Fahrenheit Scale

11. Which of the curve in the figure represents the relation between Celsius and Fahrenheit temperatures?

- a) a
b) b
c) c
d) d



Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XI 'Science'

Time: 25min.

SET B

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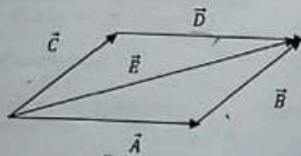
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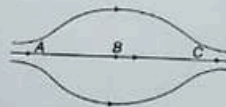
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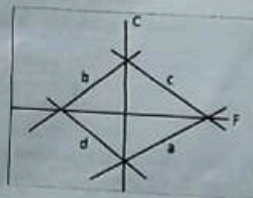
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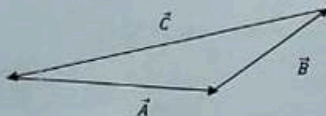
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4. What is the angle between $(\vec{P} + \vec{Q})$ and $(\vec{P} - \vec{Q})$?

- a) Zero b) π c) $\frac{\pi}{2}$ d) π

5. A force of 120 N and a force of 20 N acts simultaneously at a point may produce a resultant force of:

- a) 80 N b) 140 N c) 160 N d) 0 N

6. A certain vector in x-y plane has an x-component of 12 m and y-component of 8m. It is then, rotated in xy plane so that its x-component is halved. Then its new y component is approximately

- a) 14m b) 10.12m c) 2.14m d) 13.11m

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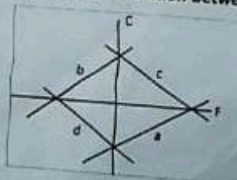
9. The magnitude of electric field strength E such that an electron placed in it would experience an electrical force equal to its weight, is given by:

- a) mg b) mg/e c) mg/e^2 d) e/mg

10. Two thermometers are constructed in such a way that one has a spherical bulb and the other has a elongated cylindrical bulb, which one will respond more quickly to temperature changes?

- a) with cylindrical bulb b) with spherical bulb
c) both respond similarly d) any one may respond quickly depending upon the value of temperature

11. Which of the curve in the figure represents the relation between Celsius and Fahrenheit temperatures?



- a) a
b) b
c) c
d) d

- b) State and explain the principle of superposition of Coulomb's force with suitable diagram. [2]
- c) A polythene piece rubbed with wool is found to have a $-ve$ charge of $0.64 \mu C$. (i) estimate the number of electrons transferred from which to which? (ii) is there a transfer of mass from wool to polythene? If yes, find the amount of mass transferred. [2]

18. a) Find the dimensions of Planck's constant. If its value in cgs system is $6.62 \times 10^{-27} \text{ erg-sec}$, what will be its value in MKS system? [2]
- b) Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon velocity v , the density ρ and acceleration due to gravity g , show that mass m varies directly as the sixth power of the velocity of the flow. [3]

Long answer questions:
GROUP 'C' [3x8=24]

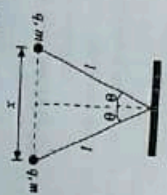
20. a) What is meant by resolution of a vector? Why resolving a vector is important in physics? Can the walking of a man be an example of resolution of vectors? Explain. [3]
- b) Can the magnitude of the rectangular component of a vector be greater than the magnitude of that vector. Explain. [3]
- c) A man rows a boat with a speed of 18 kmh^{-1} in the north-west direction. The shoreline makes an angle of 15° south of west. Find the components of velocity of the boat along the shoreline and perpendicular to the shoreline. [2]
- d) An athlete runs 2 km , east and then 3 km , 40° south of east. Find the magnitude and direction of total displacement. [2]
21. a) What do you mean by absolute zero and absolute scale of temperature? [2]
- b) A student claimed that the thermometers register their own temperature therefore, they are useless. How will you respond? [2]

- c) Mercury boils at 357°C . However, mercury thermometers are made such that they can measure temperature up to 500°C . Explain how this is possible. [1]

- d) Suppose you come across old scientific notes that describe a temperature scale called Z on which the boiling point of water is 65.0°Z and the freezing point is -14.0°Z . To what temperature on Fahrenheit scale would a temperature of $T = -98.0^\circ\text{Z}$ correspond? Assume that Z scale is linear, that is, size of a Z degree is the same everywhere on the Z-scale. [1]

22. a) Explain modern theory of electrification? [3]
- b) Define 1 Coulomb of charge on the basis of Coulomb's law. [1]
- c) When a glass rod is rubbed with a silk cloth, charges appear on both. A similar phenomenon is observed with many other pairs of bodies. Explain how this observation is consistent with the law of conservation of charge. [1]

- d) Two small spheres each of mass m and positive charge q , hang from light threads of lengths l . Each thread makes an angle of θ with the vertical as shown in the figure. [2]



- (i) On the diagram draw and label all the forces on spheres. [1]
- (ii) Develop an expression for the charge q in terms of m , l , θ and g and the Coulomb's law constant. [3]

"Best of Luck"

KATHMANDU MODEL SECONDARY SCHOOL

FIRST-TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3hrs.

Subject: Physics '1011'

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'B' [8×5=40]

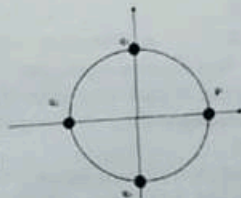
Short answer questions:

12. a) What are dimensional and non-dimensional variable? Give examples of each. [2]
- b) Check the correctness of the physical relation $Y = \frac{Mg}{\pi r^2 l}$ where Y = Young's modulus Mg is weight applied on wire, l is length of wire, r is increase in length of wire, and r is radius of wire. [2]
- c) How many times the units of energy be increased if each of the units of mass, length and time is increased by two times. [1]
13. a) Two people are pulling a heavy cart from the same point but in slightly different directions. One pulls with a force of magnitude A , the other with force B , and the angle between their ropes is θ . The cart doesn't turn—it moves straight in a direction that results from both pulls happening simultaneously. Using the rule that combines two vectors acting from the same point at an angle, derive an expression for the total force acting on the cart (its magnitude and direction). [3]
- b) The two forces are given by: $\vec{F}_1 = 500 \text{ N, east}$ and $\vec{F}_2 = 600 \text{ N, north}$. Draw a vector diagram to subtract $\vec{F}_2$ from $\vec{F}_1$. Find the magnitude and direction. [2]
14. a) A scalar may be a pure number or a physical quantity. What is the effect of multiplication of a vector by a scalar? [2]
- b) Give the geometrical interpretation of dot product of two vectors. [1]

- c) If $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = -2\hat{i} - 5 + 7\hat{k}$, find the angle between $\vec{A}$ and $\vec{B}$. [2]

15. a) The expression of electric field E due to a point charge at any point near to it is defined by $\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$, where q is the test charge and F is the force acting on it. What is the significance of $\lim_{q \rightarrow 0}$ in this expression? [1]
- b) Why two electric field lines cannot intersect with each other? [1]

- c) Consider three equal charges q_1, q_2 , and $q_3 = q = +1\mu\text{C}$ located at three different points on a circle of radius 1m , as shown in the figure. Calculate the total electric field at point P due to this distribution of charges. [3]



16. a) What are induced and inducing charges. How will you charge a conductor positively by the method of induction. Explain with clear diagram. [1+2]
- b) A small ball of paper has a mass of $9 \times 10^{-5} \text{ kg}$ and carries a charge of $5 \mu\text{C}$. When it is held over another charged ball of paper at a distance of 2 cm above it, the two balls stay in equilibrium, what is the charge on the second ball? [2]
17. a) What are electric field lines? [1]
- b) If one field line represents $1.0 \times 10^9 \text{ C}$ of charge, determine the number of field lines drawn for a charge of $2 \mu\text{C}$. [1]
- c) Two-point charges $+2 \mu\text{C}$ and $+8 \mu\text{C}$ are placed 10 cm apart in air. Sketch qualitatively the electric field lines, indicate the region and calculate the position where the resultant field will be zero. [3]
18. a) Write the conditions for validity of Coulomb's law of force between charges. [1]

Determine the magnitude of the electrostatic force on the electron at this location and state its direction. If the electron is released from rest at this location midway between the plates, determine its speed just before striking one of the plates. Assume that the gravitational effects are negligible.

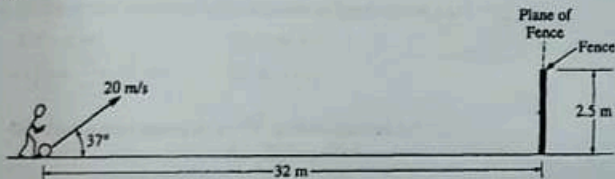
[1+2]

18. a) Derive a relation that shows the variation of density of a solid with temperature. A liquid has a density of 1000 Kg m^{-3} at 0°C . If its coefficient of linear expansion is $6.67 \times 10^{-5} / ^\circ\text{C}$, What will be its density when it heated to 50°C ? [2+1]
- b) A steel wire of diameter 2 mm is stretched between two rigid clamps with an initial tension of 100 N at a temperature of 40°C . If the temperature of the wire decreases to 10°C , calculate the new tension in the wire. (Given, $Y_{\text{steel}} = 2 \times 10^{11} \text{ N/m}^2$ and $\alpha_{\text{steel}} = 1.2 \times 10^{-5} / ^\circ\text{C}$) [2]

GROUP 'C' [3×8=24]

Long answer questions.

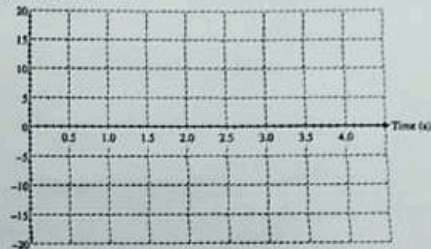
20. a) What is projectile motion. Show that with proper sketch the path followed by a horizontal projectile is parabolic in nature. [3]
- b)



A ball of mass 0.5 kilogram, initially at rest, is kicked as shown in figure.

- i) Determine the time it takes for the ball to reach the plane of the fence. [1]
- ii) Will the ball hit the fence? If so, how far below the top of the fence will it hit? If not, how far above the top of the fence will it pass? [1]
- iii) What will be the velocity when the ball is at the plane of the fence? [1]

- iv. On the axes below, sketch the horizontal and vertical components of the velocity of the ball as functions of time until the ball reaches the plane of the fence. [2]



21. a) State and explain Newton's law of cooling with its limitations. [3]
- b) In an experiment to determine the specific latent heat of vaporization of steam L_f , the following measurements were taken: mass of calorimeter + stirrer = x kg, mass of water = y kg, mass of steam condensed = z kg, initial temperature of water = $\theta_1^\circ\text{C}$, specific heat capacities of calorimeter and water are s_1 and s_2 respectively. Express L_f in terms of above data. [2]
- c) What are the difference between water equivalent and thermal capacity of a body? [1]
- d) From what height a block of ice is dropped in order that it may completely melt. It is assumed that 20% of energy of fall is retained by ice? ($L_f = 80 \text{ cal/gm}$) [2]
22. a) Define electric flux and find its dimensional formula. [2]
- b) State and prove Gauss theorem of Electrostatics. Write some properties of Gaussian surface. [1+1+1]
- c) A hollow spherical conductor of radius 12 cm has a uniform surface charge density of 13.26 nC/m^2 . Calculate the electric field strength at a distance (i) 6 cm and (ii) 4 m from the centre of the sphere. [1+2]

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SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET-B

Time: 25min.

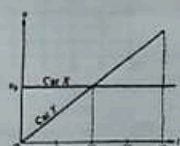
Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Choose the best alternative.

GROUP 'A' [11×1=11]

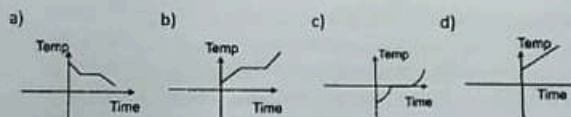
- In the expression, $\frac{1}{2} \epsilon_0 E^2$ (where ϵ_0 is permittivity and E is electric field) has same dimensions as that of:
 (a) Surface Tension (b) Force (c) Power (d) Pressure
- Given that $\vec{P} + \vec{Q} + \vec{R} = 0$, and $|\vec{P}| = 3$, $|\vec{Q}| = 5$ and $|\vec{R}| = 7$, what is the angle between $\vec{P}$ and $\vec{Q}$?
 (a) 120° (b) 60° (c) 45° (d) 30°
- A ball initially moves horizontally with velocity v_i as shown. It is then struck by a stick. After leaving the stick, the ball moves vertically with a velocity v_f , which is smaller in magnitude than v_i . Which of the following vectors best represents the direction of the average force that the stick exerts on the ball?

 (a)  (b)  (c)  (d) 
- At time $t = 0$, car X travelling with speed v_0 passes car Y which is just starting to move. Both cars then travel on two parallel lines of the same straight road. The graph of speed v versus time t for both cars is shown. Which of the following is true at time $t = 20$ seconds?

 (a) Car Y is behind car X
 (b) Car Y is passing car X
 (c) Car Y is in front of car X
 (d) Car X is accelerating faster than Car Y

- A clinical thermometer is usually much shorter than

a laboratory thermometer, because:

- It is marked in Fahrenheit not in Celsius
 - a long column in inconvenient
 - Only a limited range of temperature is required.
 - a long column of mercury cannot be introduced in the glass tube.
- Two copper rods of equal length but unequal diameter are heated through the same range of temperature. Its increase in length will be:
 (a) more for thinner rod (b) more for thicker rod
 (c) same for both the rods (d) considerable for thinner rod
 - A concave mirror with a radius of curvature of 1.0 m is used to collect light from a distant star. The distance between the mirror and the image of the star is most nearly
 (a) 0.25 m (b) 0.50 m (c) 0.75 m (d) 1.0 m
 - Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time?



- one gram hydrogen atom has 6×10^{23} atoms. Imagine that all the nuclei are put at the north pole of the earth and the electrons are at the south pole of the earth (radius 6400 km). The force between the charges is:
 (a) $1.0 \times 10^6 N$ (b) $5.0 \times 10^5 N$
 (c) $2.5 \times 10^5 N$ (d) $2.0 \times 10^5 N$
- A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in random unit associated with the curved surface B, the flux linked with the plane surface A in the same units will be:

 (a) $\frac{q}{2\epsilon_0}$ (b) $\frac{\phi}{3}$ (c) $\frac{q}{\epsilon_0} - \phi$ (d) $\frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$

- If electric field is uniform, then the electric lines of force are:

- divergent
- convergent
- circular
- parallel

Class: XI 'Science'

Time: 3hrs.

Subject: Physics '1011'

F.M.: 75

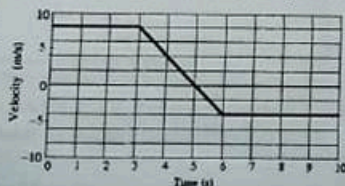
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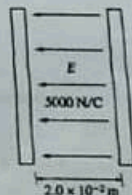
Attempt all the questions:

GROUP 'B' [8×5=40]

12. a) The volume per second of a liquid flowing through a horizontal pipe of length l is given by $\frac{kpa^3}{4\eta}$, where k is constant, p is the excess pressure (force per unit area), a is the radius of the pipe and η is a frictional quantity of dimensions $[ML^{-1}T^{-1}]$. By dimensionally find the value of x . [2]
- b) A swimmer can swim in still water with a speed of 2 m/s. He intends to cross a river 200 m wide. The river flow is non-uniform: in the first 100 m, the current speed is 1 m/s, but in the second 100 m the current increases to 1.5 m/s. Find, (i) the minimum time of crossing the river (ii) total time taken to cross the river if the swimmer reaches a point exactly opposite to his starting point [1+2]
13. a) What is instantaneous velocity? How will you find its value from the position-time graph of a moving object? Explain. [2]
- b) The velocity vs. time graph for the motion of a car on a straight track is shown in the diagram. The thick line represents the velocity. Assume that the car starts at the origin $x = 0$.
i) At which time, the car is at greatest distance from the origin? [1]
ii) What is the average velocity of the car for the 10 second interval? [2]
14. a) Derive the relation between superficial and cubical expansivity of the material. [3]



- b) Pendulum of a clock gives correct time at 20°C . How many seconds a day will it gain or lose when the temperature falls to 5°C ? ($\alpha_{\text{pendulum}} = 0.0002/^\circ\text{C}$) [2]
15. a) What is meant by relative velocity? Write the general expression to obtain the relative velocity with appropriate diagram. [2]
- b) To a man walking due east at the rate of 2 kmh^{-1} , rain appears to fall vertically. When he increases his speed to 4 kmh^{-1} , it appears to meet him at an angle of 45° . Find the real direction and speed of the rain. [3]
16. a) Derive a relation for Radius of curvature, object distance and image distance for a concave mirror when virtual image is formed? [3]
- b) An object 2cm in height is placed at 30 cm from a convex mirror of radius of curvature 60 cm. Find the position and size of image. [2]
17. a) State the principle of conservation of linear momentum. [1]
- b) Prove the principle for a system of two interacting bodies using Newton's third law of motion. [2]
- c) A ball A of mass 0.1 kg moving with a velocity of 6 m/s collides directly with a ball B of mass 0.2 kg at rest. Calculate their common velocity if both balls move off together. If A had rebounded with a velocity of 2 m/s in the opposite direction after collision, what would be the new velocity of B? [2]
18. a) What are electric field lines? How does the density of electric field lines change when they enter a dielectric medium (like glass or water) from a vacuum? [2]
- b) Two parallel conducting plates, each of area 0.30 m^2 are separated by a distance of $2.0 \times 10^{-2}\text{ m}$ of air. One plate has charge $+Q$ and another plate has charge $-Q$. An electric field of 5000 N/C is directed to the left in the space between the plates, as shown in the diagram.
i) Indicate by redrawing diagram which plate is +ve and which plate is -ve.
ii) An electron is initially located at a point midway between the plates.



KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET-A

Time: 25min.

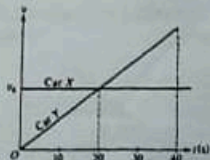
Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

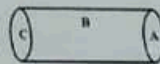
Choose the best alternative.

GROUP 'A' [11×1=11]

- In the expression, $v = At^2 + \frac{B}{C+t}$, where v is velocity and t is time, what is the dimensional formula for the product ABC ?
 (a) $[ML^2T^{-2}]$ (b) $[L^2T^{-2}]$
 (c) $[L^2T^{-3}]$ (d) $[M^0L^2T^0]$
- If the two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other, what is the value of m ?
 (a) 1 or 2 (b) -1 or 2 (c) 2 or 2 (d) 1 or -2
- A ball initially moves horizontally with velocity v_0 as shown. It is then struck by a stick. After leaving the stick, the ball moves vertically with a velocity v_1 , which is greater in magnitude than v_0 . Which of the following vectors best represents the direction of the average force that the stick exerts on the ball?
 (a) (b) (c) (d)
- At time $t = 0$, car X travelling with speed v_0 passes car Y which is just starting to move. Both cars then travel on two parallel lines of the same straight road. The graph of speed v versus time t for both cars is shown. From time $t = 0$ to time $t = 40$ seconds, the areas under both curves are equal. Therefore, which of the following is true at time $t = 40$ seconds?
 (a) Car Y is behind car X. (b) Car Y is passing car X.



- Car Y is in front of car X. (d) Car X is accelerating faster than Car Y.
- When a liquid is heated in a glass vessel, the level of liquid first falls slightly before it begins to rise. This initial fall is due to:
 (a) The evaporation of liquid (b) Contraction of liquid
 (c) The anomalous expansion of liquid (d) The expansion of glass vessel
- A bimetallic strip made of brass and iron is heated. Brass has a higher coefficient of linear expansion than iron. In which direction will the strip bend?
 (a) Towards the iron side (b) It will not bend, only increase in length
 (c) towards the brass side (d) it will twist into spiral
- Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time?
 (a) (b) (c) (d)
- A ray of light passing through the centre of curvature of a concave mirror is reflected back along the same path. Why?
 (a) The mirror is perfectly flat at that point (b) The angle of incidence is 0°
 (c) The ray is parallel to principal axis (d) The angle of incidence is 90°
- If the radius of a Gaussian sphere enclosing a charge is halved, how does the total electric flux through the surface change?
 (a) It becomes four times larger (b) It remains the same
 (c) It becomes half (d) It becomes zero
- A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in random unit associated with the curved surface B, the flux linked with the plane surface C in the same units will be:
 (a) $\frac{q}{2\epsilon_0}$ (b) $\frac{\phi}{3}$ (c) $\frac{q}{\epsilon_0} - \phi$ (d) $\frac{1}{2}(\frac{q}{\epsilon_0} - \phi)$
- Which of the following values of dielectric constant K is impossible for any material?
 (a) 80 (b) 0.5 (c) 1.0 (d) infinity





KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET-A

Time: 25min.

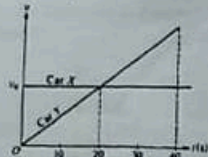
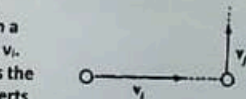
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Choose the best alternative.

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- A ball initially moves horizontally with velocity v_i as shown. It is then struck by a stick. After leaving the stick, the ball moves vertically with a velocity v_f , which is greater in magnitude than v_i . Which of the following vectors best represents the direction of the average force that the stick exerts on the ball?
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